

Random vectors and PCA

Many variables at once: a mean vector, a covariance matrix, and the few directions that hold most of the spread.

THE QUESTION

How do we summarise d numbers that vary together, and which few directions carry most of their variation?

THE INTUITION

A vector of d variables has a mean vector μ and a $d \times d$ covariance matrix Σ . The eigenvectors of Σ are the axes of the data cloud and the eigenvalues are the variances along them. PCA keeps the axes with the largest eigenvalues.

KEY TAKEAWAY

$AX + b$ has mean $A\mu + b$ and covariance $A\Sigma A^\top$. PCA: centre, eigen-decompose, project onto the top k eigenvectors; $\lambda_i / \sum_j \lambda_j$ is the share each keeps.

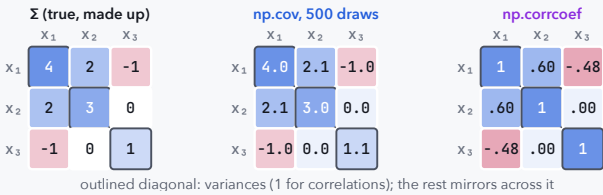
data $X: n \times d \rightarrow$ **centre** $\rightarrow S = X_c^\top X_c / (n - 1) \rightarrow$ **eigenvectors, eigenvalues** $\rightarrow$ **project on top k** $\rightarrow$ **rebuild, check the error**

1 One draw per row; the mean is a vector

```
rng = np.random.default_rng(0)
X = rng.multivariate_normal(
    [5, -2], [[1, .7], [.7, 1]], 1000)
X.shape           # (1000, 2): n x d
m = X.mean(axis=0) # [ 5.04 -1.98]
Xc = X - m         # centred
Xc.T @ Xc / (len(X) - 1)
# [[1.014 0.740]
#  [0.740 1.043]]
np.cov(X, rowvar=False) # the same
np.cov(X).shape         # (1000, 1000)!
```

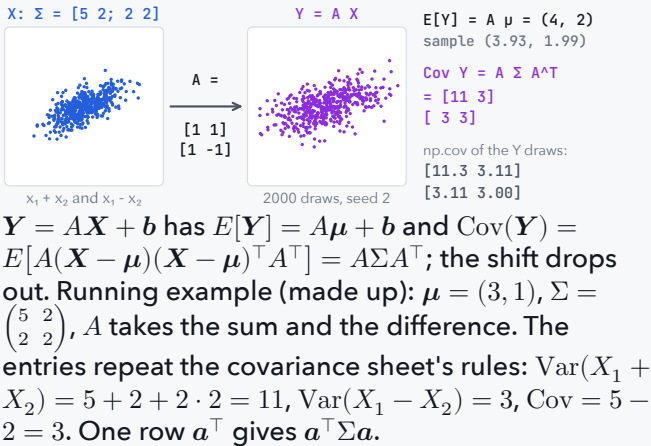
$X = (X_1, \dots, X_d)$ is d random variables observed together, with mean $\mu = (E[X_1], \dots, E[X_d])$. Data: an $n \times d$ matrix, one draw per row (the PCA notebook's normal). `np.cov` and `torch.cov` read rows as variables: pass `rowvar=False` or `X.T`. Both divide by $n - 1$; dividing by n gives 1.013.

2 The covariance matrix holds every pairwise covariance



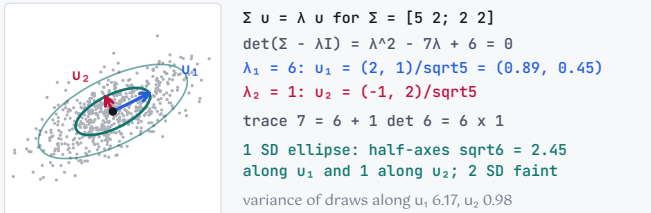
$\Sigma_{ij} = \text{Cov}(X_i, X_j)$, so $\Sigma = E[(X - \mu)(X - \mu)^\top] = E[XX^\top] - \mu\mu^\top$. It is symmetric, with the variances on the diagonal. For every a , $a^\top \Sigma a = \text{Var}(a^\top X) \geq 0$: Σ is positive semi-definite. Entry (i, j) over $\sigma_i \sigma_j$ gives the correlation matrix.

3 Linear maps give $A\mu + b$ and $A\Sigma A^\top$



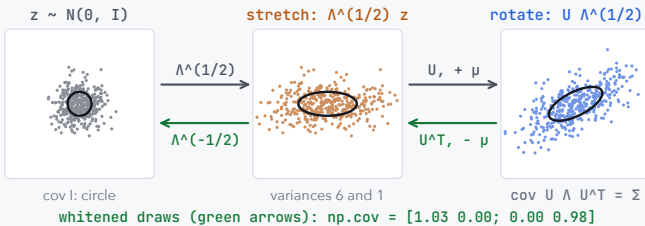
$Y = AX + b$ has $E[Y] = A\mu + b$ and $\text{Cov}(Y) = E[A(X - \mu)(X - \mu)^\top A^\top] = A\Sigma A^\top$; the shift drops out. Running example (made up): $\mu = (3, 1)$, $\Sigma = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$, A takes the sum and the difference. The entries repeat the covariance sheet's rules: $\text{Var}(X_1 + X_2) = 5 + 2 + 2 \cdot 2 = 11$, $\text{Var}(X_1 - X_2) = 3$, $\text{Cov} = 5 - 2 = 3$. One row $a^\top$ gives $a^\top \Sigma a$.

4 Eigenvectors of Σ are the axes of the cloud



A symmetric Σ has d orthonormal eigenvectors: $\Sigma = U\Lambda U^\top$ with $\lambda_i = u_i^\top \Sigma u_i = \text{Var}(u_i^\top X) \geq 0$. An eigenvalue is the variance along its eigenvector, which has length 1. The eigenvalues add up to the trace (total variance) and multiply to $|\Sigma|$. For a normal, the 1 SD ellipse has half-axes $\sqrt{\lambda_i}$ along u_i : the tilted ellipses of the covariance sheet.

5 White noise, stretched and rotated, gives any Σ ; whitening undoes it



If $Z \sim \mathcal{N}(0, I)$, then $X = \mu + U\Lambda^{1/2}Z$ has covariance $U\Lambda U^\top = \Sigma$ (the normal-shape app). It is the d -dimensional normal, with linear maps and marginals that stay normal:

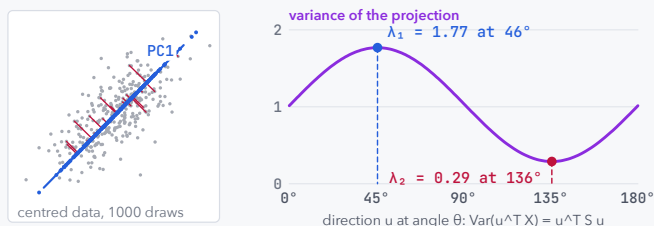
$$f(x) = \frac{\exp(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu))}{(2\pi)^{d/2} |\Sigma|^{1/2}}.$$

Whitening, $\Lambda^{-1/2}U^\top(x - \mu)$, gives covariance I . Standardising columns gives the correlation matrix (0.645 here; true $2/\sqrt{10} = 0.632$).

Principal component analysis

Rotate onto the eigenvectors of the covariance, keep the few with the most variance, and check what is lost.

6 PC1 is the direction of largest variance



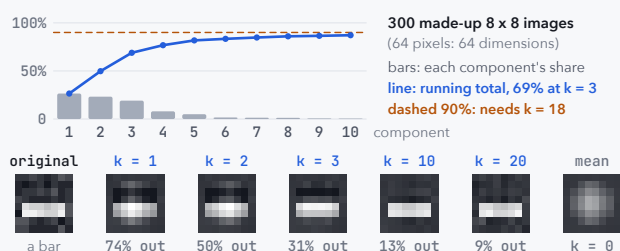
Among unit vectors, $\text{Var}(u^T X) = u^T S u$ peaks at the top eigenvector, with value λ_1 ; the best direction at right angles is u_2 , and so on. On panel 1's data, $\lambda_1 = 1.77$ at 46° and $\lambda_2 = 0.29$. Projecting keeps one score per point, $z_i = u_1^T(x_i - \bar{x})$, rebuilt as $\bar{x} + z_i u_1$ (blue). The squared red gaps average $\lambda_2(n-1)/n = 0.288$, the dropped eigenvalue.

7 PCA in a few lines of NumPy

```
S = np.cov(Xc, rowvar=False) # d x d
lam, U = np.linalg.eigh(S)    # ascending
lam, U = lam[::-1], U[:, ::-1] # big first
Z = Xc @ U[:, :k]             # scores
Xhat = Z @ U[:, :k].T + m     # rebuild
lam / lam.sum()               # [0.86 0.14]
_, s, Vt = np.linalg.svd(Xc, full_matrices=False)
s**2 / (len(X) - 1)           # equals lam
```

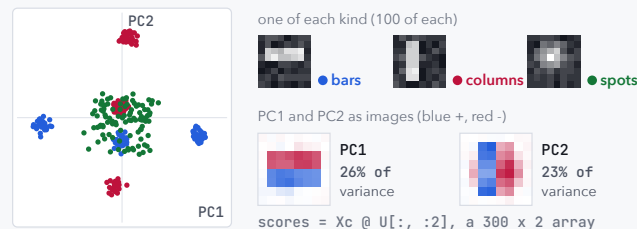
X_c and m are from panel 1. `eigh` sorts eigenvalues from small to large, so PC1 is the last column. The SVD of X_c gives the same axes (rows of V_t) without forming S . Signs are arbitrary: a library may flip a component.

8 Explained variance guides the choice of k



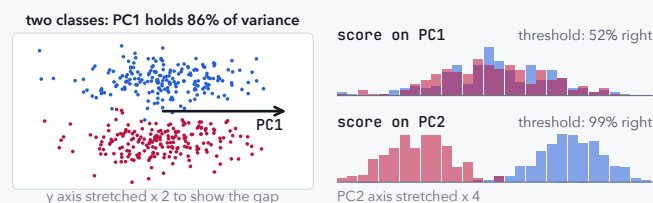
Component i keeps $\lambda_i / \sum_j \lambda_j$ of the total variance $\text{tr } \Sigma$. Here pixel noise spreads thinly over many directions, so 90% needs 18 components. Rebuilds of one bar (bottom) sharpen with k ; the mean squared error over all images, about the sum of the dropped eigenvalues, falls from 418 at $k=1$ to 73 at $k=10$. Pick k at the elbow or a target share.

9 Two scores put 64-pixel images on a page



Each image is a point in $\mathbb{R}^{64}$; its scores on PC1 and PC2 place it in a plane, as the digits app does for 8×8 handwritten digits. Here PC1 contrasts top and bottom rows, so bars sort along it by the row they sit in; PC2 does the same for columns; spots stay near the centre. The groups show because they vary in different ways. PCA never saw a label.

10 Standardise mixed units; PCA ignores labels



Units: for 500 made-up trips (km and minutes, $r = 0.81$) PC1 is (0.28, 0.96). In metres it becomes (1.00, 0.002) with 100% of the variance: the biggest numbers win. Standardised, both give (0.71, 0.71). Labels: here the two classes differ along the narrow direction. PC2 holds 14% of the variance, yet a threshold on it gets 99% right; on PC1, 52%. High variance does not mean useful for telling classes apart.

11 Pause and think

$\Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$: eigenvalues and axes? 5 along (1, 1), 3 along (1, -1).

Running example: $\text{Var}(2X_1 + X_2)$? $a^T \Sigma a = 20 + 2 + 8 = 30$.

Can $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ be a covariance matrix? No: $a = (1, -1)$ gives -2 .

Eigenvalues 6, 3, 1: share kept by two? $9/10 = 90\%$. 20 images of 100 pixels: nonzero eigenvalues? At most 19.

Every column doubled? Eigenvalues times 4; same axes and shares.

$$\text{Cov}(AX + b) = A \Sigma A^T \cdot \Sigma = U \Lambda U^T \cdot \text{PCA: centre, eigen-decompose, keep top } k \cdot \text{share kept } \lambda_i / \sum_j \lambda_j$$

Source: the course notebooks "Random vector" and "Principal component analysis" (centre, `eigh`, `project`, `rebuild`, explained variance), the digits app and the normal-shape app, nipunbatra.github.io/psdv-teaching. All data are our own and illustrative (the notebook's 2D normal redrawn with seed 0, a made-up Σ , trips and 8×8 images), computed by the sheet's figure script. This is the last sheet of the series, which starts with the arrays and tables sheet.